

Section:

L3

Q. 1: [2 + 4 = 6 points] Karachi has many tall buildings, but only some of them have a good view of the Arabian Sea. Suppose we have an array $A[1..n]$ that stores the heights of buildings from north to south in the city. If a building is taller than all other buildings to its south, then we say that it has a good view of the sea.

An algorithm that returns a stack of buildings that have a good view is shown below. The returned stack has a list of buildings (their heights) that have a good view of the sea.

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(b) Using amortized analysis, find the time complexity of the above algorithm in terms of the array size n .

[illegible]

Q. 2: [1 + 1 + 1 + 1 = 4 points] Suppose you have a Deque (double ended queue data structure). The only operations allowed on the Deque are:

- `void addToFront(x)`, `void addToRear(x)` //No element returned
- `T deleteFromFront()`, `T deleteFromRear()` //The deleted element is returned
- `size()` //returns the size of the deque as the number of elements

Use the above methods only to create new operations on the Deque as follows (you may use temporary variables if needed). Assume that you have a deque of at least size 2, i.e. there are at least 2 elements in the deque.

Write the following methods for the deque (using the above methods only)

(a) delete and return the second element of the deque i.e. `T delete_second()`.

(b) insert an element `x` at the second last position in the deque, i.e. `void ins_2_last(x)`

(c) Use your methods in (a) and (b) to create a new method, i.e. `void move( )` that moves (shifts) the second element in the deque to the second last element in the deque.

(d) If the input deque is [b, o, w, l, s] what will be the deque if `move()` is applied two times?

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